

ORION Score: A Calibrated Composite Metric for Large Language Model Citation Probability

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ORION Score: A Calibrated Composite Metric for Large Language Model Citation Probability

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Abstract

The rise of generative AI search engines — ChatGPT, Claude, Perplexity, Gemini — has introduced a measurement problem that existing SEO authority scores do not solve. Metrics such as Domain Authority, Domain Rating, and Authority Score are trained against SERP rank. They do not directly predict the outcome that matters

on the retrieval-augmented answer surface: whether a given web property will be *cited* inside a generated response. This paper introduces the **ORION Score**, the first open, decomposable, and probabilistically calibrated composite metric purpose-built for that outcome. ORION aggregates five weighted pillars — Originality, Retrievability, Information Extractability, Observed Visibility, and Notability (whose initials spell the score’s name) — each grounded in a specific stage of the retrieval-augmented generation pipeline. The composite is a 0-100-scaled probability, decomposable into pillar sub-scores that map to concrete interventions; the weight distribution is dynamic, redistributing toward Observed Visibility when live citation evidence is available. Three design commitments distinguish ORION from the existing commercial authority landscape: (i) an **open methodology** released under CC BY 4.0, (ii) a **probabilistic calibration target** rather than a rank proxy, and (iii) an explicit **Honesty Gate** — a code-enforced research-release flag that does not flip to “validated” until a set of pre-declared auditable thresholds is met on a reproducible dataset. ORION is presented as a shared vocabulary for Generative Engine Optimisation (GEO) measurement, complementing rather than replacing the SERP-targeted authority scores.

Keywords: Generative Engine Optimisation, GEO, large language model, retrieval-augmented generation, citation probability, content authority, ORION Score.

1. Introduction

1.1 A new retrieval surface

Between 2023 and 2026, web search underwent its largest structural change since the transition to mobile. A rising share of queries — conservatively estimated at 25% in early 2026 — is now mediated by large language models (LLMs) that synthesise answers from retrieved sources rather than returning a ranked list of links (Lewis et al., 2020; Aggarwal et al., 2024). Four engines dominate the consumer experience of this surface: OpenAI’s ChatGPT, Anthropic’s Claude, Perplexity, and Google’s Gemini. Each is a commercial application of the retrieval-augmented generation (RAG) pattern, with a proprietary retriever, a re-ranker, and a generator.

1.2 A measurement gap

The existing SEO measurement stack was not built for this substrate. Three widely adopted commercial scores — Moz Domain Authority (Moz, 2023), Ahrefs Domain Rating (Ahrefs, 2023), Semrush Authority Score (Semrush, 2022) — are trained predominantly against SERP-rank outcomes. They are link-graph metrics with secondary traffic or SERP-feature signals. They predict Google ranking potential reasonably well. They do not predict the outcome that now matters for a growing share of traffic: whether a source is *cited inside a generated answer*.

The two outcomes are related but not interchangeable. A domain with Domain Authority 85 may never appear in a ChatGPT response for its core category because its content lacks the extractability signals that RAG retrievers preferentially surface. Conversely, a domain with Domain Authority 30 and strong passage-level extractability may appear in the majority of relevant LLM answers. Empirically, the correlation

between a high SERP-rank authority score and the observed citation rate on LLM answer engines is moderate at best — the SERP-oriented stack cannot be retargeted; its feature set and its training target are both wrong for the new task.

1.3 Contribution

This paper contributes the ORION Score: a 0-100 composite metric whose design target is, explicitly, **citation probability inside LLM-generated answers**. The contribution is threefold.

First, **architectural**: ORION is the first published, reproducible, five-pillar decomposition in which each pillar is grounded in a specific stage of the RAG pipeline — index inclusion, retrieval, chunk-level extraction, re-ranking, and post-retrieval trust filtering — rather than an intuitive content-quality rubric. The decomposition is actionable: a practitioner reading “ORION: 42, pillar R: 20” can identify exactly which stage of the RAG pipeline their property is failing.

Second, **probabilistic**: the composite is calibrated so that a score of 70 is intended to represent a 0.70 probability of citation on the queries for which the page is the most relevant candidate. This is a stronger claim than the rank-proxy target of DA/DR/AS and requires an explicit calibration procedure, which this paper specifies.

Third, **honesty-gated**: because the calibration target is probabilistic, the paper commits to a pre-declared validation protocol with quantitative thresholds. Until those thresholds are met on a reproducible dataset, the system self-reports `is_research_release: true` in every API response. This transforms “validated” from a marketing label into an engineering state with an auditable transition event. No widely deployed SEO authority score in the literature operates under such a commitment.

1.4 Organisation

Section 2 situates ORION in prior work across information retrieval, retrieval-augmented generation, quality frameworks, and commercial authority scoring. Section 3 formalises the methodology and derives the pillar weights from retrieval-mechanistic reasoning. Section 4 differentiates ORION from adjacent scores. Section 5 specifies the validation protocol and the Honesty Gate. Section 6 acknowledges limitations. Section 7 lists future work. Section 8 concludes. A formal reference list follows.

2. Related Work

2.1 Information retrieval foundations

Modern web ranking traces to the vector-space model (Salton, Wong, & Yang, 1975), which formalised lexical similarity between a query and a document, and to the probabilistic relevance framework including BM25 and its successors (Robertson & Zaragoza, 2009). Authority-based ranking was introduced by PageRank (Brin & Page,

1998), which extended pure lexical relevance with a link-graph prior. The dominant SEO authority scores — Moz DA, Ahrefs DR, Semrush AS — are descendants of that link-graph lineage with proprietary modern refinements.

2.2 Retrieval-augmented generation

The methodological bridge from search to LLM citation was established by Lewis et al. (2020). RAG decouples parametric knowledge (what the model learned from its training corpus) from non-parametric knowledge (what it can retrieve at inference time) and introduces a two-stage pipeline: retrieve a candidate passage set from an index, then generate a response conditioned on those passages. Commercial answer engines are applications of this pattern with proprietary retrievers, re-rankers, and grounding strategies.

The practical consequence for content strategy is that the retrieval unit is typically a *chunk* — a contiguous passage of ~300 tokens — rather than a whole page. A content property with strong page-level authority but weak chunk-level self-containment can fail the retrieval stage entirely and therefore never appear in an LLM answer regardless of its topical relevance or link profile. This is a central mechanistic claim on which ORION’s Information Extractability pillar rests.

2.3 Quality frameworks

Google’s Search Quality Evaluator Guidelines (Google, 2024) formalised the E-E-A-T rubric (Experience, Expertise, Authoritativeness, Trustworthiness), which has become a shared vocabulary for content quality. The Helpful Content Update (Google, 2022) operationalised several of these signals at the algorithmic layer. ORION incorporates E-E-A-T-derived signals additively into the Originality and Notability pillars, rather than treating E-E-A-T as a separate multiplier; this removes the double-counting problem present in earlier versions of ORION and in several commercial composite scores.

2.4 Commercial authority scoring systems

Three commercial scores dominate the SEO tooling market:

- **Moz Domain Authority** — a regression model trained to predict SERP rank, with inputs dominated by link-graph features (Moz, 2023).
- **Ahrefs Domain Rating** — a log-scale metric derived primarily from the linking-domain profile (Ahrefs, 2023).
- **Semrush Authority Score** — a hybrid of link profile, organic traffic estimation, and SERP feature presence (Semrush, 2022).

All three share a structural limitation for the LLM-citation task: the training target is SERP rank, and the input features are dominated by signals that do not map onto the RAG retrieval pipeline. They are also all closed methodologies; the published documentation describes inputs at a high level but does not permit independent reproduction or auditing.

2.5 Entity and brand signals

Work by Barnard (2020) on branded-query SERP composition, and Google’s own Knowledge Graph announcement (Singhal, 2012), established that entity-level signals — brand mention density, entity co-occurrence, knowledge-panel presence — affect surface composition independently of link authority. LLMs rely heavily on entity priors during retrieval and generation: a source associated with a well-resolved entity in the pretraining corpus is disproportionately likely to appear in the model’s candidate set. These findings anchor ORION’s Notability pillar.

2.6 Generative Engine Optimisation

Aggarwal et al. (2024) introduced *Generative Engine Optimization* as a research topic in their arXiv paper of the same title (arXiv:2311.09735). They presented the first systematic empirical study of how page-level content interventions — adding citations, adding statistics, direct quotation, fluency modifications — affect LLM-generated answers, and reported measurable uplift from specific interventions. Their study establishes GEO as a distinct optimisation surface from SEO but does not propose a composite score for the optimisation.

Practitioner work by Neil Patel, Aleyda Solis, Marie Haynes, and Lily Ray has subsequently extended the applied conversation, and category coinages such as *Answer Engine Optimization* (AEO) and *LLM SEO* have proliferated. None of this body of work proposes a published, reproducible composite metric targeting citation probability directly. **ORION’s contribution sits in that specific gap.**

3. The ORION Methodology

3.1 Design principles

ORION is designed around five commitments:

1. **Decomposability.** The composite must break into pillars that map to specific interventions. A practitioner reading “ORION: 42” must be able to ask “which pillar is dragging me down?” and obtain an answer that corresponds to a concrete optimisation action.
2. **Mechanistic grounding.** Each pillar must correspond to a specific, identifiable stage in the RAG retrieval pipeline, not to an intuitive content-quality rubric.
3. **Probabilistic calibration.** The composite is a calibrated probability of citation, not a rank proxy. Scores are reported on the 0–100 scale but interpreted as $100 \times P(\text{cite})$.
4. **Dynamic adaptation.** When live citation evidence exists for a property, the weight distribution redistributes toward Observed Visibility so that empirical evidence dominates the composite.
5. **Honesty under uncertainty.** Empirical claims are gated behind a pre-declared validation protocol. The score reports its own epistemic state.

3.2 Formal definition

Let p denote a web page and $Q(p)$ denote the distribution of queries for which p is topically relevant. Let $c(q, p)$ denote the binary event that p is cited in a generated response to query q from a target LLM engine. The underlying quantity ORION estimates is:

$$\text{cite-rate}(p) = \mathbb{E}_{q \sim Q(p)} [c(q, p)]$$

That is, the expected citation rate of page p over its topically relevant query distribution. The ORION Score is a computable proxy for this expectation constructed from five pillar sub-scores.

3.3 Pillar definitions, mechanism, and weight derivation

Each pillar corresponds to a specific stage of the RAG pipeline. Weights are not arbitrary; they reflect the elasticity of the overall citation probability with respect to each stage, estimated from practitioner literature and refined against live probe data as it accumulates.

O₁ — Originality • weight 0.20 *RAG stage:* retrieval-index redundancy reduction. A RAG retriever operates over a corpus in which many pages express overlapping claims. The retriever has an implicit de-duplication pressure — it prefers to surface passages that introduce information not already redundantly represented in the candidate set. Originality therefore controls the *marginal retrieval probability* of a page conditional on the candidate set composition.

Operational inputs: embedding-distance from the nearest-neighbour competitor content for the same topic; detection of first-party data, proprietary research, or novel analytical frameworks; earliest-publication signal on the substantive claim set, with a citation-backflow adjustment; perspective novelty scored against the dominant framing via stance detection.

Why this weight: for non-incumbent brands, Originality is the single largest lever under their direct control. The 0.20 weight reflects a middle estimate between practitioner estimates of 0.25–0.35 (which overstate, by conflating Originality with Information Extractability) and the ~ 0.12 observed in Aggarwal et al.’s (2024) content-intervention study (which understates because their ablation held everything else constant, leaving no room for compound effects).

R — Retrievalability • weight 0.15 *RAG stage:* index inclusion and parseability. A page that is not in the target engine’s retrieval index cannot be cited regardless of its quality. A page that is in the index but whose structure defeats the parser — unresolvable heading hierarchy, broken schema, Core Web Vitals failures that cause render-time crawl abandonment — receives a reduced parse quality score that attenuates all downstream retrieval.

Operational inputs: presence in Google and Bing search indexes as a proxy for RAG-retriever inclusion; schema markup coverage across Article, FAQPage, HowTo, Person,

and Organization types; heading hierarchy integrity; Core Web Vitals (LCP, INP, CLS); and the absence of exclusion directives (noindex, llm-no-train).

Why this weight: Retrievability is a necessary gate, not a differentiator above a threshold. A page with Retrievability 90 and one with Retrievability 100 will both be retrieved; the 10-point gap does not translate to a proportional citation-probability gap. The 0.15 weight reflects gate-like behaviour: below ~40, Retrievability dominates the composite; above ~70, it saturates.

I — Information Extractability · weight 0.30 *RAG stage:* chunk-level retrieval within the candidate set. Once the candidate set is assembled, the retriever operates at the passage level. A ~300-token chunk extracted from the page must retain semantic coherence, package claims self-containedly, and present entity-dense, answer-ready structure. Two pages with identical page-level topical coverage can produce dramatically different chunk-level retrieval outcomes.

Operational inputs: answer-readiness (density of question-form headings immediately followed by direct, self-contained answer passages); factual density (count of specific claims — dates, numbers, named entities, quantified comparisons — per 100 words); structural clarity (definition blocks, comparison tables, step-by-step lists); entity density weighted by entity specificity; chunk self-containment (whether a ~300-token chunk extracted at any position retains semantic coherence without surrounding context).

Why this weight: Information Extractability is the largest weight because chunk-level extractability is the single strongest lever on whether a RAG retriever surfaces a passage. Aggarwal et al. (2024) report the largest measured uplifts in their intervention study on factors that map to this pillar (citations, statistics, direct quotations). The 0.30 weight reflects that empirical finding.

O₂ — Observed Visibility · weight 0.25 (static) / 0.35 (dynamic) *RAG stage:* posterior over actual citation events. When live probe data is available for a property, Observed Visibility is the closest thing ORION has to a ground-truth signal. It is measured by running a probe set of N queries (N = 100–500 per vertical) against ChatGPT, Claude, Perplexity, and Gemini and recording: (i) citation rate, (ii) response position when cited, (iii) cross-engine consistency.

Operational inputs: citation rate across probe queries; response position (first-cited sources receive higher weight); cross-engine consistency (uniform citation across engines is rewarded; single-engine dominance is partially discounted).

Why the dynamic weight: when direct evidence of the citation outcome is available, prediction should defer to evidence. The dynamic redistribution raises O₂ from 0.25 to 0.35 and rescales the other four pillars proportionally:

$$O_1 : 0.17, \quad R : 0.13, \quad I : 0.25, \quad O_2 : 0.35, \quad N : 0.10$$

This is formally a Bayesian-style update: as evidence accumulates, the composite moves from a prior (structural signals) toward a posterior (observed outcomes).

N — Notability · weight 0.10 *RAG stage:* re-ranker entity prior. LLMs use entity recognition as a retrieval-time prior. Sources associated with well-resolved, cross-corpus-consistent entities (Wikipedia, Wikidata, knowledge-panel-present brands) are preferentially surfaced because the re-ranker has higher confidence the source’s claims are verifiable.

Operational inputs: brand mention density across a rolling web-crawl sample; Wikipedia and Wikidata presence and depth; curated-directory presence (G2 for SaaS, IMDb for entertainment, Crunchbase for startups, etc.); co-occurrence strength with other recognised entities in the same topical neighbourhood.

Why this weight: Notability provides a stable, persistent boost — but one that is slow to move. The 0.10 weight reflects that stability: Notability rarely changes the composite direction for a given intervention cycle.

3.4 Composite formula

The static ORION Score is the weighted sum of the five pillar sub-scores, each normalised to [0, 100]:

$$\text{ORION}_{\text{static}}(p) = 0.20 \cdot O_1 + 0.15 \cdot R + 0.30 \cdot I + 0.25 \cdot O_2 + 0.10 \cdot N$$

When live Observed Visibility data is available, the weights redistribute:

$$\text{ORION}_{\text{dynamic}}(p) = 0.17 \cdot O_1 + 0.13 \cdot R + 0.25 \cdot I + 0.35 \cdot O_2 + 0.10 \cdot N$$

The composite is then scaled to [0, 100] and interpreted as $100 \times P(\text{cite})$ over the topically relevant query distribution.

3.5 Score bands

Band	Range	Interpretation
Minimal	0-19	Largely invisible to AI. Fundamental improvements required.
Low	20-39	Limited AI presence. Significant optimisation needed.
Moderate	40-59	Some AI visibility. Clear opportunities to improve.
Good	60-79	Strong AI presence. Regular appearances in AI answers.
Excellent	80-100	Go-to source. Appears in most relevant AI answers.

Bands are calibrated against the observed empirical distribution once validation data is available.

3.6 The Citability Gap diagnostic

ORION reports two score variants per property when both are computable:

- **Predictive ORION** — the composite computed without live probe data, using only the structural pillars O_1 , R , I , N with O_2 estimated from static signals.

- **Observed ORION** — the composite computed with live probe data, where O_2 is empirical.

The **Citability Gap** is defined as:

$$\text{Gap}(p) = \text{Predictive}(p) - \text{Observed}(p)$$

A positive Gap indicates a page that the model expects to be citable but that underperforms empirically — often a signal of a trust issue, a platform-specific retrieval exclusion, or a competitive displacement event. A negative Gap indicates a page outperforming its structural profile — a learning signal about which structural features are currently under-weighted, and occasionally a flag that manual-authority tailwinds are compensating for a structural weakness.

The Gap is not a defect of the model; it is the model’s most direct diagnostic output. A world in which Predictive and Observed always agree is a world in which no learning is possible.

4. Differentiation

Dimension	ORION	Moz DA	Ahrefs DR	Semrush AS
Primary target	LLM citation probability	Google SERP rank	Google SERP rank	Google SERP rank
Time domain	Near-real-time with O_2	Monthly	Weekly	Weekly
Link-graph features	Low weight	High	Dominant	High
Content / structural features	High weight	Low	Low	Moderate
Entity features	Medium (N)	None	None	Low
Live LLM probes as first-class signal	Yes (O_2)	No	No	No
Open methodology	Yes (CC BY 4.0)	Partial	Partial	Partial
Probabilistic calibration	Yes (P(cite))	No (rank proxy)	No	No
Pre-declared validation thresholds	Yes (§5)	No	No	No

ORION is the only score in the current landscape whose primary target is LLM citation, whose methodology is fully open, whose calibration target is an explicit probabil-

ity, and whose empirical claims are gated behind a pre-declared validation protocol. For a site that cares about both Google visibility and AI visibility, running DA/DR/AS alongside ORION is recommended — the two sets of scores answer different questions.

5. Validation Protocol and the Honesty Gate

5.1 Engineering state, not marketing claim

ORION v5.1 ships as a *research release*. This is not a softening adjective; it is an engineering state enforced in code by a boolean flag `is_research_release` wired directly to the system’s calibration version. The flag is `true` in every API response at the time of publication. It transitions to `false` only when a set of pre-declared, auditable thresholds is met on a reproducible dataset.

5.2 Threshold specification

Before `is_research_release` can flip to `false`, all of the following must hold on a held-out, temporally-separated test set:

Dataset scale. - $\geq 2,000$ labelled training rows drawn from a page frame of $\geq 3,000$ pages under citation-propensity stratification, with ≥ 300 positive-label rows. - ≥ 6 verticals represented in the training set; no single vertical exceeding 30% of rows. - All four target platforms (ChatGPT, Claude, Perplexity, Gemini) represented by ≥ 300 rows each.

Temporal generalisation. - A held-out test set of ≥ 500 rows with a temporal cutoff at least 30 days after the training-set cutoff.

Performance. - $\text{PR-AUC} \geq 0.50$ - $\text{ROC-AUC} \geq 0.70$ - Expected Calibration Error ≤ 0.08 over 10 bins - Top-quintile lift $\geq 2.0\times$ significant at $p < 0.01$

5.3 Transition event

When the thresholds are met, a reproducible validation report is published, containing dataset metadata, model version, coefficient or feature-importance summary, calibration plot, lift table by quintile, and per-vertical / per-platform breakdown. The flag flips in a single auditable event. Prior to that event, no claim of “validated ORION” is made, in any channel.

This is the **Honesty Gate**, and it is the most consequential single commitment in this methodology.

6. Limitations

Probe query sampling. Observed Visibility is measured against a finite probe set. The recommended size ($N = 200$ per vertical) is a cost/variance compromise; it is not

a theoretical optimum.

LLM retriever and re-ranker drift. Target engines update their retrievers and generators frequently. An ORION score measured in one quarter is not strictly comparable to one measured in a later quarter. Methodology version and measurement date are therefore reported with every score.

Originality measurement sensitivity. The embedding-distance originality signal depends on the choice of embedding model. ORION standardises on a single public embedding model per methodology version to preserve comparability within a version.

Long-tail Notability under-representation. Small sites in niche verticals may score artificially low on Notability because brand mention density falls below the detection floor of the web-crawl sample. A minimum-support threshold is applied before N is penalised.

Downstream business outcomes. ORION predicts citation frequency. Citation is a necessary condition for the downstream commercial outcomes — branded traffic, referral visits, revenue — but not a sufficient one. ORION should be read alongside those outcome metrics, not as a substitute for them.

Observability asymmetry across engines. Citations are exposed prominently by some engines (Perplexity), inconsistently by others (ChatGPT, Claude), and rarely in default modes by some (Gemini). Observed Visibility is therefore systematically noisier for citation-opaque engines, and the score should be read with awareness of this per-engine variance.

7. Future Work

Open probe dataset. Release the full probe query set and probe response corpus under a permissive licence to enable independent validation.

Community-contributed probes. Accept probe submissions from vertical specialists, subject to a review protocol to preserve query-set quality.

Model-specific sub-scores. Expose ORION-ChatGPT, ORION-Claude, ORION-Perplexity, and ORION-Gemini as separate sub-scores, for teams whose audience concentrates on a specific engine.

Multi-language validation. The current v5.1 methodology validates on English-language content. A planned v6.0 will extend validation to the top ten languages.

Longitudinal drift study. Re-score the reference dataset quarterly to measure methodology stability and LLM-retrieval drift over time.

Chunk-level API surface. Expose ORION at the chunk resolution in addition to the page resolution, with per-chunk pillar sub-scores and an identified “best-performing chunk” for each page.

8. Conclusion

The ORION Score is offered as the first open, calibrated, composite probability metric for LLM citation. By aggregating Originality, Retrievability, Information Extractability, Observed Visibility, and Notability into a single weighted score — with weights derived from the mechanisms of the retrieval-augmented generation pipeline, with dynamic redistribution when live evidence is available, and with empirical claims gated behind a pre-declared Honesty threshold — the methodology provides a shared vocabulary for GEO measurement that parallels what Domain Authority, Domain Rating, and Authority Score have provided for traditional SEO, while measuring a fundamentally different outcome.

The methodology is released openly, under CC BY 4.0, and versioned, on the expectation that it will evolve as LLM retrieval infrastructure evolves. This paper is the definitional anchor for version 5.1.

References

1. Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., & Deshpande, A. (2024). *GEO: Generative Engine Optimization*. arXiv:2311.09735. <https://arxiv.org/abs/2311.09735>
2. Ahrefs. (2023). *Domain Rating (DR): What It Is & How to Improve It*. <https://ahrefs.com/blog/domain-rating/>
3. Barnard, J. (2020). *Brand SERP: A New Paradigm for Brand Communication on Google*. Kalicube Publications.
4. Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1-7), 107-117.
5. Google. (2022). *August 2022 Helpful Content Update*. <https://developers.google.com/search/content-update>
6. Google. (2024). *Search Quality Evaluator Guidelines*. <https://static.googleusercontent.com/m>
7. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.
8. Moz. (2023). *Domain Authority 2.0: Methodology Overview*. <https://moz.com/learn/seo/domain-authority>
9. Robertson, S., & Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4), 333-389.
10. Salton, G., Wong, A., & Yang, C. S. (1975). A vector space model for automatic indexing. *Communications of the ACM*, 18(11), 613-620.
11. Semrush. (2022). *Authority Score: Methodology*. <https://www.semrush.com/blog/authority-score/>
12. Singhal, A. (2012). *Introducing the Knowledge Graph: Things, Not Strings*. Google Official Blog. <https://blog.google/products/search/introducing-knowledge-graph-things-not/>

Appendix A: Pillar Sub-Score Computation Reference

Each pillar sub-score is computed as a weighted combination of its operational inputs (see §3.3), each normalised to $[0, 100]$. Formal sub-score formulae are specified in the accompanying engineering documentation and are version-stamped to the methodology version. Appendix A is held for that version-stamped specification in a subsequent release.

Appendix B: Probe Query Protocol

The probe query set for a property consists of 100–500 natural-language queries generated via a two-stage procedure: (1) seed queries extracted from the property’s topical vector; (2) fan-out via paraphrase and intent-cluster expansion. Queries are balanced across intent types (informational, comparison, how-to). Temperature and prompt-structure conventions for the target engines are version-stamped.

Appendix C: Methodology Changelog

- **v3** (2025): Four-pillar composite; static weights.
- **v4** (Jan 2026): Renamed to ORION; five-pillar decomposition; structural scoring only.
- **v5.0** (Jan 2026): Added Observed Visibility; introduced static/dynamic weight redistribution; first public release.
- **v5.1** (Apr 2026): Formalised the RAG-stage mechanism per pillar; derived weights from retrieval elasticity; introduced the Honesty Gate with quantitative thresholds; added formal References section; published under CC BY 4.0.

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